

02425 Diffusions and Stochastic Differential Equations

Exercises Week 5: The Euler Method and the Ito Integral

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In the following, we use the following Python implementation of the Itô integral;

```
def ito_integral(G, B):  
    return np.concatenate(([0], np.cumsum(G[:-1] * np.diff(B))))
```

Question 1 Verify the implementation by computing and plotting

$$\int_0^T \cos(t) d\sin(t)$$

for $T \in [0, 2\pi]$ on a sufficiently fine grid (at least 100 grid points).
Establish the analytical result

$$\int_0^T \cos(t) d\sin(t) = \frac{1}{2}T + \frac{1}{4}\sin(2T)$$

and compare with the numerical result. *Hint:* use $d\sin t = \cos t dt$ and $\cos^2(t) = \frac{1}{2} + \frac{1}{2}\cos(2t)$;

We establish the analytical result with the two rules given by the hint; (1) $d\sin(t) = \cos(t)dt$ and (2) $\cos^2(t) = \frac{1}{2} + \frac{1}{2}\cos(2t)$. From (1) it follows;

$$\int_0^T \cos(t) d\sin(t) = \int_0^T \cos^2(t) dt$$

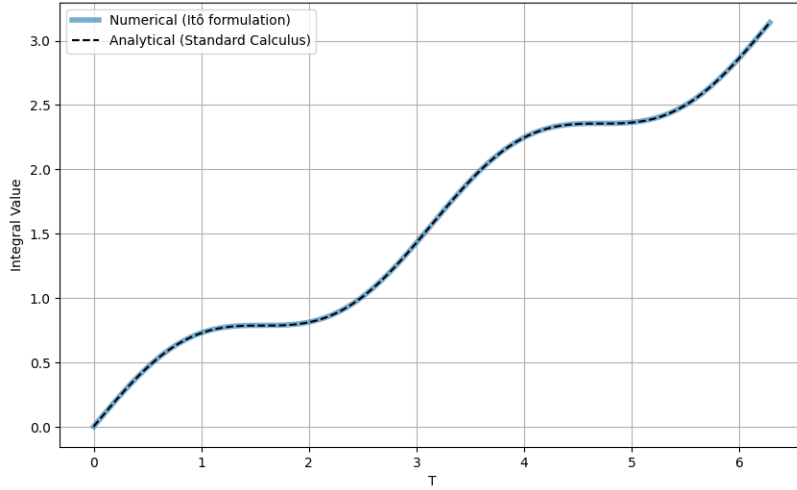
and from (2) we then have

$$\int_0^T \cos^2(t) dt = \int_0^T \left(\frac{1}{2} + \frac{1}{2}\cos(2t) \right) dt$$

This is solved as

$$\left[\frac{1}{2}t + \frac{1}{4}\sin(2t) \right]_0^T = \frac{1}{2}T + \frac{1}{4}\sin(2T)$$

We evaluate both solutions and plot them to verify the implementation;



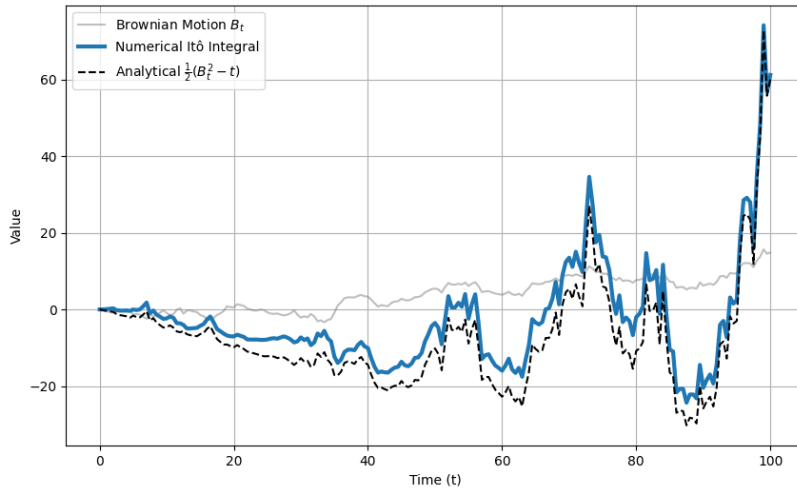
Integrating Brownian Motion wrt Itself

Question 2 Re-create figure 6.6 in the notes (page 128). Specifically, apply your integrating function to compute on realization of the Itô integral

$$I_t = \int_0^t B_s dB_s$$

for $t \in \{0.0, 0.5, 1.0, \dots, 100\}$, where $\{B_s : 0 \leq s \leq 100\}$ is Brownian motion. Plot the result as function of t . Compare with the analytical result $I_t = \frac{1}{2}(B_t^2 - t)$;

Since an Itô integral has the form $\int_0^T G_s dB_s$, we set $G = B$ for this Itô integral to compute it with our established function `itointegral()`. Comparing that the plot of this over $T = 100$ with a stepping size of 0.5 to the analytical solution yields;



Note that the stepping size is relatively large, which makes our numerical solution less precise. While the curves for both the numerical and analytical solution follow the same pattern, it is possible to see how the numerical solution is not 100% precisely on top of the analytical all the time. When the steps are this large, it opens up for overshooting/undershooting the actual regression for each step which is why we see the numerical solution oscillating around the analytical.

Question 3 Write a modified integrator which computes the "right hand approximation" given by (for general integrand f and integrator g)

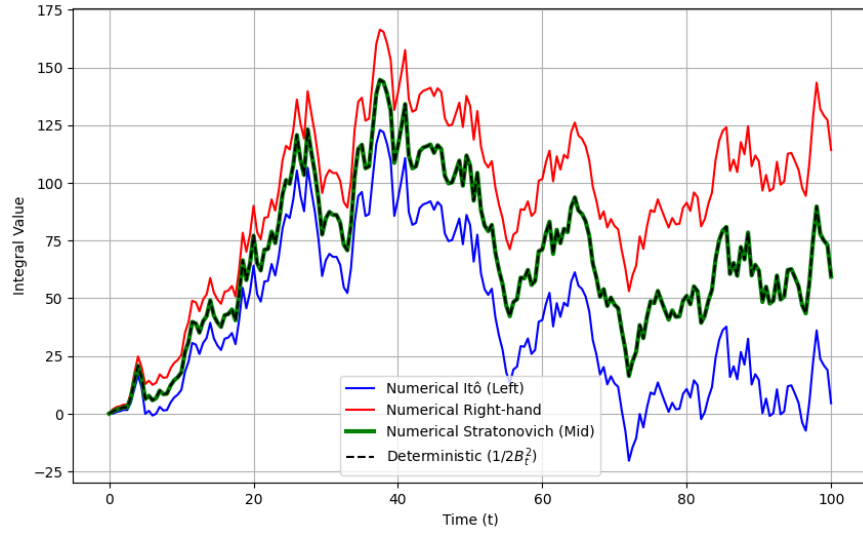
$$I_m^R = \sum_{i=1}^m f(t_i) \cdot [g(t_i) - g(t_{i-1})]$$

Add this integral to the plot. Furthermore, add the Stratonovich approximation

$$I_m^S = \sum_{i=1}^m \frac{1}{2} (f(t_{i-1}) + f(t_i)) \cdot [g(t_i) - g(t_{i-1})]$$

Finally, add the result that one could expect from deterministic calculus, i.e. $\frac{1}{2}B_t^2$;

We plot the three numerical solutions next to the deterministic;



The Stratonovich line sits on top of the deterministic solution, because it corresponds to evaluating at the midpoint thereby neutralizing the small fluctuations over time, making it evaluate to $\frac{1}{2}B_t^2$.

The Itô line therefore systematically undershoots because it accumulates the negative penalty term $-\frac{1}{2}t$. The right-hand line instead overshoots by the same positive penalty term $\frac{1}{2}t$. We notice that the gap between the lines increase as t grows as a result of this. Concretely the gap between the right-hand solution and Itô solution for $t = 100$ is thereby also exactly 100.

Question 4 (Compare exercise 6.9 in the book) Consider the Itô integral

$$I_t = \int_0^t B_s dB_s = \frac{1}{2}B_t^2 - \frac{1}{2}t$$

Use the properties of Gaussian variables (exercise 3.13) to determine the mean and variance of I_t , and verify that this agrees with the properties of Itô isometry;

We start determining the mean by applying the linearity of expectancy

$$\mathbb{E}[I_t] = \frac{1}{2}\mathbb{E}[B_t^2] - \frac{1}{2}t$$

We have the standard variance formulated as $\mathbb{V}[B_t] = \mathbb{E}[B_t^2] - (\mathbb{E}[B_t])^2$, where we know that the variance of Brownian motion at time t is equal to t , and the mean is $\mathbb{E}[B_t] = 0$. This gives us $t = \mathbb{E}[B_t^2] - (0)^2$, which we plug in;

$$\mathbb{E}[I_t] = \frac{1}{2}t - \frac{1}{2} = 0$$

With the mean established we write up the same standard variance formula for the full Itô integral;

$$\mathbb{V}[I_t] = \mathbb{E}[I_t^2] - (\mathbb{E}[I_t])^2 = \mathbb{E}[I_t^2]$$

Note that we established $\mathbb{E}[I_t] = 0$, such that we can eliminate the second term directly. We find the mean of the Itô integral squared by squaring it;

$$\begin{aligned} I_t^2 &= \left(\frac{1}{2}B_t^2 - \frac{1}{2}t \right)^2 = \left(\frac{1}{2}B_t^2 \right)^2 + \left(\frac{1}{2}t \right)^2 - 2 \left(\frac{1}{2}B_t^2 \right) \left(\frac{1}{2}t \right) \\ &= \frac{1}{4}B_t^4 + \frac{1}{4}t^2 - \frac{1}{2}B_t^2t \end{aligned}$$

We formulate the mean of this as

$$\mathbb{E}[I_t^2] = \frac{1}{4}\mathbb{E}[B_t^4] + \frac{1}{4}t^2 - \frac{1}{2}\mathbb{E}[B_t^2]t$$

By now we know $\mathbb{E}[B_t^2] = t$, and by the Gaussian variable properties we get that for any normally distributed variable like Brownian motion with mean 0 and variance t , then $\mathbb{E}[B_t^4] = 3t^2$;

$$\begin{aligned} \mathbb{E}[I_t^2] &= \frac{1}{4}3t^2 + \frac{1}{4}t^2 - \frac{1}{2}t^2 \\ &= \left(\frac{3}{4} + \frac{1}{4} - \frac{2}{4} \right) t^2 \\ &= \frac{1}{2}t^2 \end{aligned}$$

The Itô isometry theorem dictates that the expected variance of an Itô integral $\int_0^t G_s dB_s$ is equal to the standard time integral $\int_0^t \mathbb{E}[G_s^2] ds$. For our integral, $\int_0^t B_s dB_s$, this becomes

$$\int_0^t \mathbb{E}[B_s^2] ds = \int_0^t s ds = \left[\frac{1}{2}s^2 \right]_0^t = \frac{1}{2}t^2$$

Thus, we have successfully verified the Itô isometry for this integral.

Simulating an SDE with the Euler-Maruyama Method

Consider the Cox-Ingersoll-Ross process (also termed the square root process)

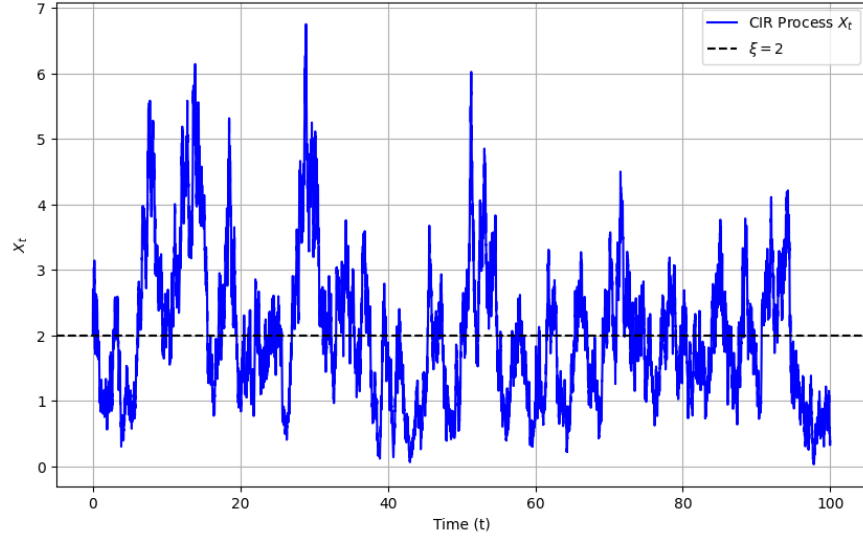
$$dX_t = \lambda(\xi - X_t) dt + \gamma\sqrt{|X_t|} dB_t$$

Question 5 Simulate the process using the Euler-Maruyama method. Take parameters $\lambda = \frac{1}{2}$, $\xi = 2$, $\gamma = 1$. Use an initial condition of $X_0 = \xi$ and simulate the process on the time interval $[0, 100]$ with a time step of $h = 0.01$. Plot the solution $\{X_t\}$ versus time t ;

We formulate the update step as

$$X_{t+h} = X_t + \lambda(\xi - X_t) + \gamma\sqrt{|X_t|}\Delta B$$

and simulate the full path;

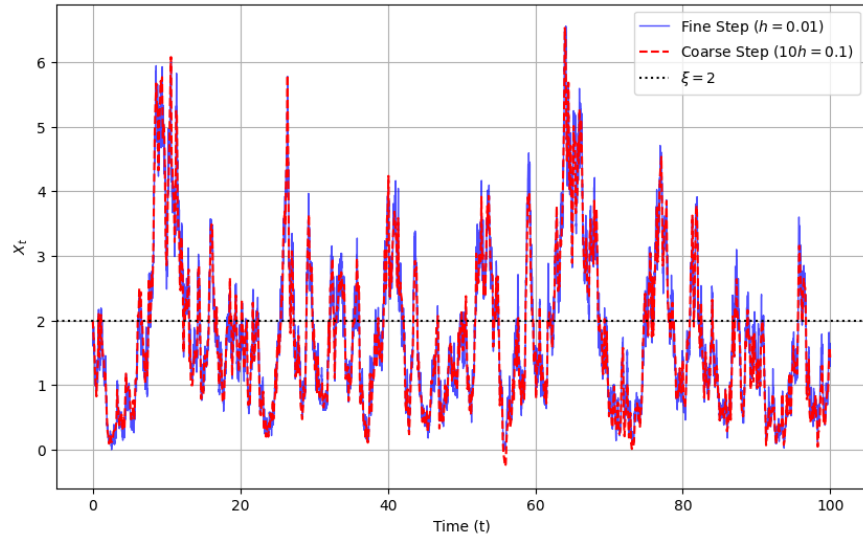


We see that the deterministic term pulls the process back towards the baseline, $\xi = 2$. From the noise term we also notice that this is larger the larger X_t is.

Effect of the Time Step

Question 6 Repeat the simulation with the same sample path of Brownian motion, but now using a time step of $10h$. Verify that both the long term and the short term behaviour is roughly the same;

We generate a sample path and sample every 10th element from it to represent the *coarse* path at $10h$;



Short term we verify that both paths are identical, as the coarse line follows the fine perfectly but misses out on small vibrations which occur for $< 0.1t$. Long term we see the two paths exhibit identical traits. We can therefore conclude that for standard SDEs, increasing the step size by factor 10 still preserves the core properties (and reduces computational overhead!).

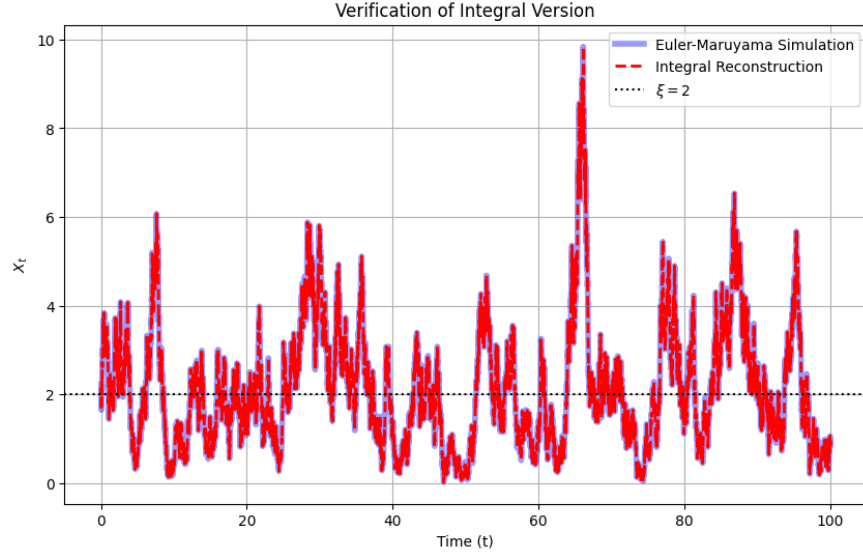
Verification of the Integral Version

Question 7 Verify numerically that the simulated sample paths of $\{B_t\}$ and $\{X_t\}$ satisfy the integral version

$$X_t = X_0 + \int_0^t \lambda(\xi - X_s) ds + \int_0^t \gamma \sqrt{|X_s|} dB_s$$

for each $t = 0.00, 0.01, 0.02, \dots, 100$. Here, you use the numerical integration routine that you made in question 1, and the verification should be by plotting the sum of the integrals on top of the Euler solution;

We reconstruct the integral by using our function `ito_integral()` defined in question 1 for both the deterministic and stochastic term;



The result confirms that the Euler-Maruyama stepping method results in the same as our integral reconstruction.

Mean, Variance, and Autocovariance Function

Question 8 Solve exercise 6.12 in the book, where we use the Euler-Maruyama method to reach ODEs for the mean, variance, and autocovariance function of the Cox-Ingersoll-Ross process;

To find the ODE for the mean, we take the Euler-Maruyama update formula for this equation;

$$X_{t+h} - X_t = \lambda(\xi - X_t)h + \gamma\sqrt{X_t}\Delta B$$

To find the continuous rate of change for the mean, we take the expected value of both sides. Since $\mathbb{E}[\Delta B] = 0$, the stochastic term vanishes;

$$\mathbb{E}[X_{t+h} - X_t] = \mathbb{E}[\lambda(\xi - X_t)h]$$

By the linearity of expectation we pass the expectation operator inside of the deterministic term;

$$\mathbb{E}[X_{t+h}] - \mathbb{E}[X_t] = \lambda(\xi - \mathbb{E}[X_t])h$$

We use the notation $m(t) = \mathbb{E}[X_t]$ for convenience;

$$m(t+h) - m(t) = \lambda(\xi - m(t))h$$

By dividing both sides by h , and letting $h \rightarrow 0$, we get the definition of the ODE of the mean;

$$\frac{dm}{dt} = \lambda(\xi - m(t))$$

To find the variance, $V(t) = \mathbb{E}[(X_t - m(t))^2]$, we see how the squared deviation changes over a time step h . We have the Euler-Maruyama updates for both terms;

$$X_{t+h} - m(t+h) = (X_t - m(t))(1 - \lambda h) + \gamma \sqrt{X_t} \Delta B$$

When we square this term, ΔB eliminates the cross product, such that we get

$$V(t+h) = \mathbb{E}[(X_t - m(t))^2(1 - \lambda h)^2] + \mathbb{E}[(\gamma \sqrt{X_t} \Delta B)^2]$$

For the second term we distribute the square $\mathbb{E}[\gamma^2 X_t (\Delta B)^2]$, where $\mathbb{E}[X_t] = m(t)$, and the variance of standard Brownian motion is defined $\mathbb{E}[(\Delta B)^2] = h$. So the second term simplifies to $\gamma^2 m(t)h$. In the first term $(1 - \lambda h)^2$ is deterministic, and can be pulled out of the expectation operator, leaving us with $\mathbb{E}[(X_t - m(t))^2]$. This is our variance, $V(t)$. Thus, we get

$$V(t+h) = V(t)(1 - 2\lambda h + \lambda^2 h^2) + \gamma^2 m(t)h$$

If we move $V(t)$ to the left side and divide by h , the ODE becomes;

$$\frac{dV}{dt} = -2\lambda V(t) + V(t)\lambda^2 h + \gamma^2 m(t)$$

Since we take the limit $h \rightarrow 0$, the term $V(t)\lambda^2 h$ approaches 0, thus yielding;

$$\frac{dV}{dt} = -2\lambda V(t) + \gamma^2 m(t)$$

To find the autocovariance function $C(s, t) = \mathbb{E}[(X_s - m(s))(X_t - m(t))]$ assuming $s < t$, we examine how the covariance evolves wrt the later time t . We multiply the state deviation at the earlier time s by the discrete Euler-Maruyama update for the deviation at time t ;

$$(X_s - m(s))(X_{t+h} - m(t+h)) = (X_s - m(s)) \left[(X_t - m(t))(1 - \lambda h) + \gamma \sqrt{X_t} \Delta B \right]$$

Expanding this expression gives;

$$(X_s - m(s))(X_{t+h} - m(t+h)) = (X_s - m(s))(X_t - m(t))(1 - \lambda h) + (X_s - m(s))\gamma \sqrt{X_t} \Delta B$$

We then apply the expectation operator to both sides to find $C(s, t+h)$. Because $s < t$, the future Brownian increment ΔB (which occurs between t and $t+h$) is independent of both the past state X_s and the current state X_t . Since $\mathbb{E}[\Delta B] = 0$, the expectation of the entire stochastic cross product vanishes;

$$C(s, t+h) = \mathbb{E}[(X_s - m(s))(X_t - m(t))](1 - \lambda h) + 0$$

Substituting the definition of $C(s, t)$ back into the equation;

$$C(s, t+h) = C(s, t)(1 - \lambda h)$$

If we distribute $C(s, t)$, move it to the left side, and divide by h , we get the difference quotient;

$$\frac{C(s, t+h) - C(s, t)}{h} = -\lambda C(s, t)$$

Taking the continuous limit as $h \rightarrow 0$ gives the final ODE for the autocovariance function with respect to t ;

$$\frac{\partial C(s, t)}{\partial t} = -\lambda C(s, t)$$

Note: the initial condition for this ODE at $t = s$ is just the variance, $C(s, s) = V(s)$.